350mA, 20V - 40V Schottky Barrier Diode

FEATURES

- AEC-Q101 qualified
- Low power loss, high current capability, low VF
- Surface mound device type
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

APPLICATIONS

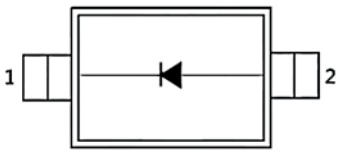
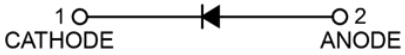
- Low voltage rectifications
- High-efficiency DC-DC conversions
- Switch mode power supplies
- Inverse polarity protections

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	350	mA
V_{RRM}	20 - 40	V
I_{FSM}	2	A
$T_{J\ MAX}$	125	°C
V_F at I_F=200mA	600	mV
Configuration	Single die	

Pb-Free RoHS COMPLIANT HALOGEN FREE

MECHANICAL DATA

- Case: SOD-123
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 10.97mg (approximately)

PACKAGE: SOD-123	PIN CONFIGURATION	CIRCUIT DIAGRAM
		

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

PARAMETER		SYMBOL	VALUE	UNIT
Power dissipation(1)		P_D	300	mW
Repetitive peak reverse voltage	SD103AWH	V_{RRM}	40	V
	SD103BWH		30	
	SD103CWH		20	
Repetitive voltage	SD103AWH	V_R	28	V
	SD103BWH		21	
	SD103CWH		14	
Forward current		I_F	350	mA

Non-repetitive peak forward surge current	t = 8.3ms	I_{FSM}	2	A
Junction temperature		T_J	-55 to +125	°C
Storage temperature		T_{STG}	-55 to +150	°C

Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance(1)	$R_{\Theta JA}$	333	°C/W

Thermal Performance Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS		SYMBOL	MIN	TYP	MAX	UNIT
Forward voltage(1)	I_F = 20mA		V_F	-	-	370	mV
	I_F = 200mA			-	-	600	
Reverse breakdown voltage	SD103AWH	I_R = 100μA	V_{BR}	40	-	-	V
	SD103BWH			30	-	-	
	SD103CWH			20	-	-	
Reverse current(2)	SD103AWH	V_R = 30V	I_R	-	-	5	μA
	SD103BWH	V_R = 20V					
	SD103CWH	V_R = 10V					
Junction capacitance	f = 1MHz, V_R = 0V		C_J	-	-	50	pF
Reverse recovery time	I_F = I_R = 200mA, I_{RR} = 20mA, R_L = 100Ω		t_{rr}	-	10	-	ns

Notes:

1. Pulse test with PW=0.3 ms
2. Pulse test with PW=30 ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING	DEVICE MARKING
SD103AWH RHG	SOD-123	3,000 / 7” Tape & Reel	S4
SD103BWH RHG	SOD-123	3,000 / 7” Tape & Reel	S5
SD103CWH RHG	SOD-123	3,000 / 7” Tape & Reel	S6

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Power Dissipation Curve

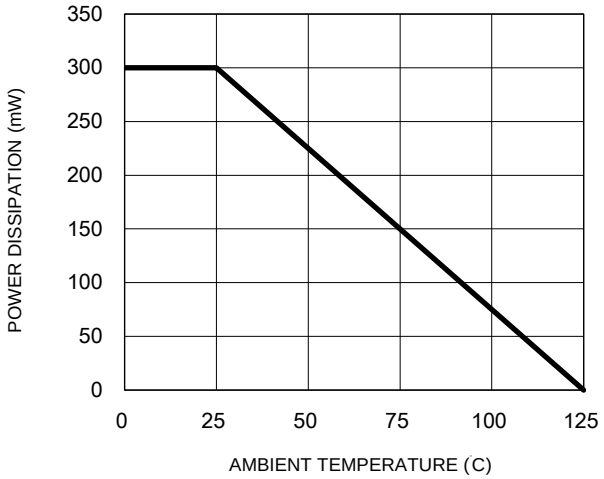


Fig.2 Typical Junction Capacitance

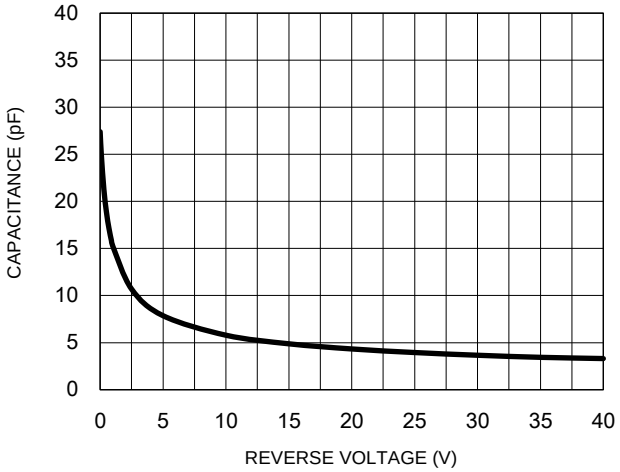


Fig.3 Typical Reverse Characteristics

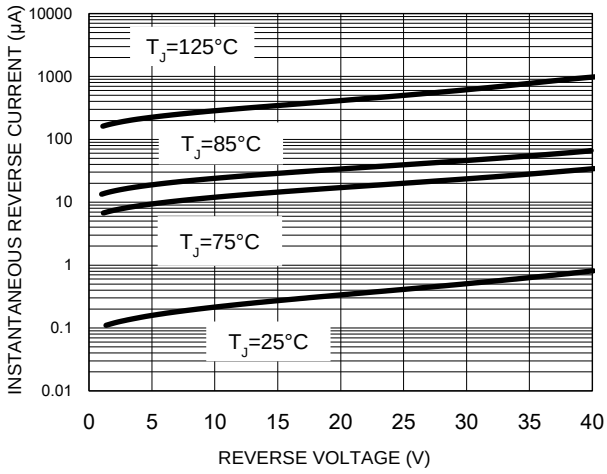
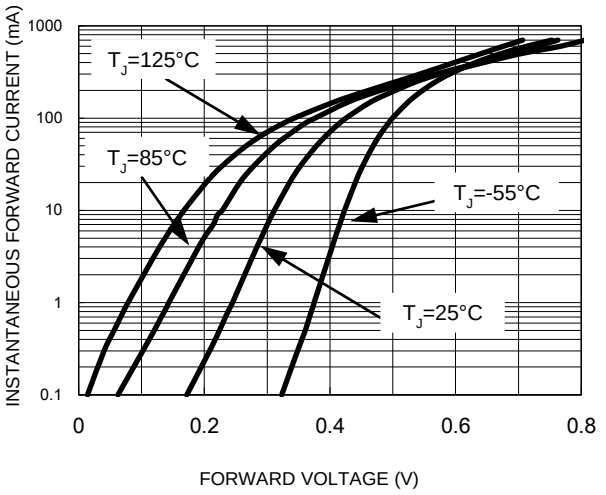
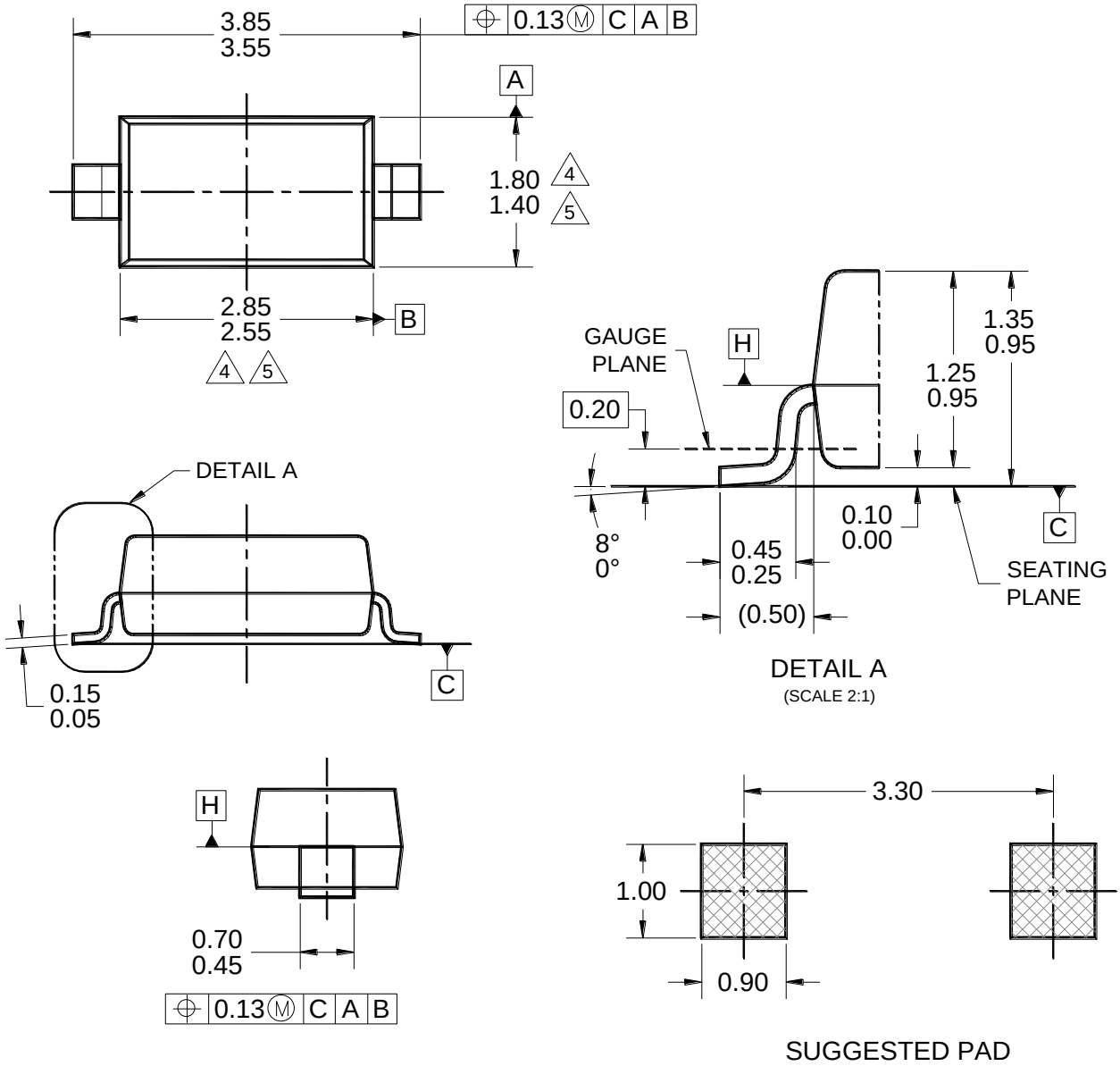


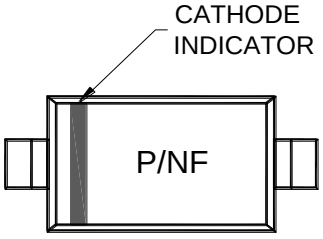
Fig.4 Typical Forward Characteristics



PACKAGE OUTLINE DIMENSION

SOD-123





P/N = Device marking
F = Factory code

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: JEDEC DO-215, VARIATION AD, ISSUE D.

4 MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH.

5 MOLDED PLASTIC BODY LATERAL DIMENSIONS TO BE DETERMINED AT DATUM PLANE H.

6. DWG NO. REF: HQ2SD07-SOD123-046 REV A.

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